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Data Analytics & Business Intelligence - Task 1

Data Cleaning, Exploratory Data Analysis (EDA) & Basic Business Reporting

Objective

The objective of Task-1 is to introduce interns to the **core foundation of data analytics**, which includes:

- Understanding raw business data
- Cleaning and transforming datasets
- Performing exploratory data analysis (EDA)
- Generating meaningful insights
- Creating basic dashboards using Excel and Power BI / Tableau
- Writing basic SQL queries

This task focuses on **data understanding before advanced analytics**.

Why This Task Matters (Industry Context)

In real organizations:

- 70–80% of analytics work involves cleaning and preparing data
- Business decisions depend on clear reports
- Analysts must convert raw data into insights

Before predictive analytics or advanced BI, professionals must:

- Identify data inconsistencies
- Handle missing values
- Perform structured analysis
- Present findings visually

Task-1 builds that essential foundation.

What You Will Build

You will work with a **Sales Dataset** and perform:

1. Data Cleaning (Excel)
2. Exploratory Data Analysis (Excel)
3. Basic SQL Queries (Data Extraction)
4. Interactive Dashboard (Power BI / Tableau)
5. Insight Report (Business Interpretation)

Dataset

Use a publicly available dataset such as:

Superstore Sales Dataset

Fields typically include:

- Order ID
- Order Date
- Customer Name
- Region
- Product Category
- Sales
- Quantity
- Profit
- Discount

(Intern may use Kaggle Superstore dataset or similar structured business dataset.)

Tools Required

- Microsoft Excel
- MySQL / SQLite / PostgreSQL
- Power BI or Tableau (Free version)
- Basic SQL interface (MySQL Workbench / DB Browser)

Step-by-Step Implementation

Part A: Data Cleaning Using Excel

Step 1: Inspect Raw Data

Check for:

- Missing values
- Duplicate rows
- Incorrect data formats
- Blank cells
- Inconsistent naming (e.g., region spelling)

Step 2: Clean the Data

Perform:

- Remove duplicates
- Replace missing values (if logical)
- Convert dates to proper format
- Standardize text values (e.g., East, WEST, west → West)
- Ensure numeric columns are properly formatted

Save cleaned dataset as:

`cleaned_sales_data.xlsx`

Part B: Exploratory Data Analysis (EDA) in Excel

Step 3: Create Summary Metrics

Calculate:

- Total Sales
- Total Profit
- Total Orders
- Average Order Value
- Total Quantity Sold

Use:

- SUM
- AVERAGE
- COUNT
- Pivot Tables

Step 4: Analyze Business Dimensions

Using Pivot Tables, analyze:

1. Sales by Region
2. Profit by Category
3. Sales by Month
4. Top 10 Customers
5. Top 5 Products

Document insights clearly.

Part C: SQL Data Extraction

Step 5: Import Data into SQL Database

Create table:

```sql
CREATE TABLE sales (
    order_id VARCHAR(50),
    order_date DATE,
    customer_name VARCHAR(100),
    region VARCHAR(50),
    category VARCHAR(50),
    sales DECIMAL(10,2),
    quantity INT,
    profit DECIMAL(10,2),
    discount DECIMAL(5,2)
);
```

Import cleaned dataset.

Step 6: Write Basic SQL Queries

Write queries to:

1. Total sales by region

```sql
SELECT region, SUM(sales) AS total_sales
FROM sales
GROUP BY region;
```

2. Top 5 profitable products

```sql
SELECT category, SUM(profit) AS total_profit
FROM sales
GROUP BY category
ORDER BY total_profit DESC
LIMIT 5;
```

3. Monthly sales trend

```sql
SELECT MONTH(order_date) AS month,
       SUM(sales) AS total_sales
FROM sales
GROUP BY month
ORDER BY month;
```

4. Highest discount impact

```sql
SELECT discount, AVG(profit)
FROM sales
GROUP BY discount;
```

Part D: Power BI / Tableau Dashboard

Step 7: Import Cleaned Data into Power BI or Tableau

Create visualizations:

- KPI Cards (Total Sales, Profit, Orders)
- Bar Chart (Sales by Region)
- Line Chart (Monthly Sales Trend)
- Pie Chart (Category Distribution)
- Top Customers Table

Step 8: Add Filters / Slicers

Include:

- Region filter
- Category filter
- Date filter

Make dashboard interactive.

Expected Output

At the end of Task-1, you must have:

- Cleaned dataset
- Pivot-based Excel analysis
- 4–5 SQL queries executed
- Business dashboard (Power BI / Tableau)
- Business insights report

Business Insights Report (Mandatory)

Prepare a 1–2 page summary including:

- Which region generates highest sales?
- Which category is most profitable?
- Are discounts affecting profit?
- Which month performs best?
- Key recommendations for management

Key Learning Outcomes

After Task-1, interns will be able to:

- Clean business datasets
- Use Excel for structured analysis
- Write SQL aggregation queries
- Create interactive BI dashboards
- Interpret business data professionally

Deliverables

Mandatory:

1. Cleaned Excel file
2. SQL query file (.sql)
3. Power BI (.pbix) or Tableau file
4. Dashboard screenshot
5. Business insight report (PDF)

Optional:

- Advanced pivot tables
- Data dictionary document

1. Excel for Data Cleaning & Analysis

Excel Basics for Data Analysis

Excel Full Course for Beginners
https://www.youtube.com/watch?v=Vl0H-qTclOg
Excel Data Cleaning Techniques
https://www.youtube.com/watch?v=4mKp8JqkXfQ
Pivot Tables Explained Clearly
https://www.youtube.com/watch?v=UsdedFoTA68
Excel Formulas for Business Analysis (SUM, COUNT, IF, VLOOKUP, XLOOKUP)
https://www.youtube.com/watch?v=ShBTJrdioLo
Working with Dates in Excel
https://www.youtube.com/watch?v=omw5Y3q5d3k
Official Microsoft Excel Help
https://support.microsoft.com/excel

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2. SQL for Data Analytics

SQL Basics (Must Learn First)

SQL Full Course (Beginner Friendly)
https://www.youtube.com/watch?v=HXV3zeQKqGY
SQL Tutorial for Data Analysis
https://www.youtube.com/watch?v=7S_tz1z_5bA

SQL SELECT, GROUP BY, ORDER BY Explained
https://www.youtube.com/watch?v=9Pzj7Aj25lw
SQL Aggregate Functions (SUM, AVG, COUNT)
https://www.youtube.com/watch?v=RSlqWnP-Dy8

SQL Practice Platforms (Free)
SQL Practice Website
https://www.sql-practice.com/
SQLite Online
https://sqliteonline.com/
MySQL Installation Guide
https://dev.mysql.com/doc/

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3. Power BI (Free Version)
Power BI Desktop (Free Download)
https://powerbi.microsoft.com/desktop/
Power BI Beginner Full Course
https://www.youtube.com/watch?v=AGrl-H87pRU
Power BI Dashboard Tutorial
https://www.youtube.com/watch?v=3oC3m-2Vb1k
Power BI Data Modeling Basics
https://www.youtube.com/watch?v=0a2oHbeSx8E
Official Power BI Documentation
https://learn.microsoft.com/power-bi/

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4. Tableau (Free Alternative)
Tableau Public (Free)
https://public.tableau.com/
Tableau Beginner Tutorial
https://www.youtube.com/watch?v=2P9hE0XjQ2Q
Tableau Dashboard Tutorial
https://www.youtube.com/watch?v=7QqKjQ6nFJY
Official Tableau Training Videos
https://www.tableau.com/learn/training

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5. Business Analysis & Insight Writing
How to Write Data Insights
https://www.youtube.com/watch?v=7zQ6cM8yQmE
Business Dashboard Best Practices
https://www.youtube.com/watch?v=V6LQpYxkF8U
KPI Dashboard Design Principles
https://www.youtube.com/watch?v=Qd8z8QYp4xE

6. Sample Datasets (Free)

Superstore Dataset (Search & Download)

https://www.kaggle.com/datasets

Google Dataset Search

https://datasetsearch.research.google.com/

Data.gov (Public Datasets)

https://www.data.gov/

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7. End-to-End Data Analytics Projects (Practice)

Excel + SQL + Power BI Project Tutorial

https://www.youtube.com/watch?v=GfXFmVfSg5M

Complete Data Analyst Project Walkthrough

https://www.youtube.com/watch?v=qfyynHBFOsM

Sales Dashboard Project Example

https://www.youtube.com/watch?v=Z2zWf9v1nZQ

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8. Community Support (Free)

Stack Overflow (SQL + Excel)

https://stackoverflow.com/

Reddit Data Analysis

https://www.reddit.com/r/dataanalysis/

Power BI Community

https://community.powerbi.com/

Tableau Community

https://community.tableau.com/

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9. Recommended Learning Path for Task-1

Day 1

Excel cleaning and formatting

Pivot tables

Day 2

SQL basics

Aggregation queries

Day 3

Power BI or Tableau basics

Import dataset

Day 4

Dashboard creation

Add filters and KPIs

Day 5

Insight writing

Final documentation

10. Essentials Checklist for Task-1

You must clearly understand:

- Data cleaning process
- Pivot table analysis
- SQL GROUP BY and aggregation
- KPI identification
- Dashboard visualization principles
- Writing business insights clearly